

George Ma

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EDUCATION

University of California, San Diego

La Jolla, CA

B.S. in Computer Science; GPA: 3.93

Expected June 2028

Relevant Courses: Design & Analysis of Algorithms, Advanced Data Structures, Data Structures & Algorithms, Systems Programming, Database Systems, Object-Oriented Programming, Computer Organization, Data Science Theory

TECHNICAL SKILLS

Languages: Go, Python, Java, C/C++, SQL, TypeScript, JavaScript, Bash/Shell, HTML, CSS, ARM Assembly

Libraries/Frameworks: FastAPI, Flask, Node.js, Express.js, LiteLLM, PyTorch, scikit-learn, React, React Native

Cloud/Infra: AWS, GCP, Kubernetes, Docker/Podman, Terraform, Jenkins, ArgoCD, Kafka, PostgreSQL/MySQL, Git

EXPERIENCE

Capital One

McLean, VA

Incoming Software Engineer Intern – DevOps/SRE (Enterprise Platforms Tech)

June 2026 – August 2026

IBM

San Jose, CA

Software Engineer Intern – Automation and AI (IBM Bob)

February 2026 – May 2026

- Built a React Native/Expo mobile AI assistant, optimizing chat rendering using FlatList for **7k+ internal users**
- Engineered **OAuth/OIDC** gateways in **Go** via PKCE, JWT/Redis-backed sessions, and **OPA-based RBAC**
- Architected a FastAPI-based **LiteLLM proxy** to route custom TypeScript agent workloads across AWS Bedrock, IBM WatsonX, and Mistral AI, reducing chat inference pipeline latency by **25%** through semantic model routing
- Automated **OpenShift (K8s)** deployment for backend, agent, and auth microservices using Terraform/Jenkins
- Optimized Strimzi-managed **Kafka** pipeline latency by **32%** by diagnosing bottlenecks with Prometheus/Grafana

Praxie AI

San Francisco, CA

Software Engineer Intern – Mobile and Backend Infrastructure

April 2025 – February 2026

- Migrated backend infra from Firebase to a containerized Express REST API on **Google Cloud Run**, provisioning services with **Terraform** and securing GitHub Actions CI/CD pipelines using Workload Identity Federation
- Architected scalable tournament and multi-player management features in React Native via **PostGIS** geo-indexing and TanStack Query for optimized client-side caching, achieving **sub-1s search latency** for 300+ active users
- Developed a fault-tolerant **NoSQL to PostgreSQL** (CloudSQL) migration pipeline by using node-pg-migrate to refactor schemas and backfill production data, improving query performance by **20%** through denormalization

Alpha Kappa Psi @ UC San Diego

La Jolla, CA

Lead Full-Stack Engineer

December 2024 – January 2026

- Spearheaded a full-stack website migration from Wix to a **Next.js**/Supabase architecture by implementing **SSR**, optimized data fetching, and reusable content workflows, improving page load times by **60%** for **2,000+ MAU**
- Led Git-based sprints via feature branches, documentation, and reviews, improving long-term maintainability
- Designed a normalized **PostgreSQL** schema serving dynamic content with **sub-1s latency** under concurrent load

PROJECTS

Neural eBPF | Go, Cilium/eBPF, Linux, Kubernetes, LangGraph, Redis, React

- Built a **self-healing** SRE system using a Go daemon to detect CPU/FD leaks via eBPF `sched_switch` tracepoints
- Engineered a Redis-backed **LangGraph agent** to automate userspace mitigations, cutting recovery to **under 10s**
- Deployed a Minikube HTTP/SSE observability stack via cgroup/debugfs mounts, reducing manual triage by **70%**

Semantic LB | Go, FastAPI, LiteLLM, Ollama, Prometheus, Kubernetes, Next.js

- Implemented an intent-aware **L7 load balancer** to route API requests to Go worker pools with a **50ms** overhead
- Designed a FastAPI/**LiteLLM classifier** with Ollama Mistral and **LRU cache** to achieve a **63% cache hit** rate
- Deployed a Minikube stack with Prometheus to display routing decisions, latency, cache hits, and worker behavior

Shadow Infra | Go, FastAPI, Kubernetes, Docker, LangGraph, Supabase, React

- Architected a **PR-triggered shadow deployment** testing system using FastAPI/GitHub webhooks for automated creation/teardown of ephemeral Kubernetes services, catching **87%** of seeded API regressions before merging PRs
- Engineered an async **Go tee proxy** to mirror traffic to production and shadow services with no user-facing delay
- Optimized a **LangGraph drift-analysis** pipeline with fast-path checks, cutting manual regression review by **80%**